LLMs as Content Curators: A Large-Scale Audit of Recommendation Bias Across Providers and Platforms

Nicolò Pagan[1], Christopher Barrie[2], Christopher A. Bail[3], Petter Törnberg[4]
[1]University of Zürich, [2]New York University, [3]Duke University, [4]University of Amsterdam

COLM

University of Zurich

NYU

Duke

University of Amsterdam

MOTIVATION

Platforms are using more and more LLM-based ranking systems.

- X moved its entire ranking system to Grok, over 100M posts a day
- Instagram introduced AI-driven Reels recommendation summaries
- Bluesky launched Attie, an agentic feed builder powered by Claude

OPEN QUESTION

Which biases are structural across providers, and which can prompting fix?

AUDITING PIPELINE

For each condition: 100 trials. Each trial samples 100 posts from a pool of 5,000, and the LLM returns the top 10. Temperature 0.3. **Models see only the post text: no metadata, no author information.**

PROVIDERS

- OpenAI GPT-4o-mini
- Anthropic Claude Sonnet 4.5
- Google Gemini 2.0 Flash

PLATFORMS

- Twitter/X
- Bluesky
- Reddit

PROMPT STYLES

- Neutral
- General
- Popular
- Engaging
- Informative
- Controversial

10 POST FEATURES (all platforms)

- Polarizing topic score
- Primary topic
- Toxicity
- Sentiment polarity
- Subjectivity
- Avg word length
- Has URL
- Has emoji
- Has hashtag
- Has mention

3 AUTHOR FEATURES (Twitter/X only, inferred from bios)

- Political leaning
- Gender
- Minority status

54 conditions × 100 trials × 100 posts = 540,000 posts ranked

Limitations. Demographics are inferred by an LLM, not verified. The polarizing topic score is a topic-based proxy, not a measure of stance. Recommendations are non-personalized.

RESULTS

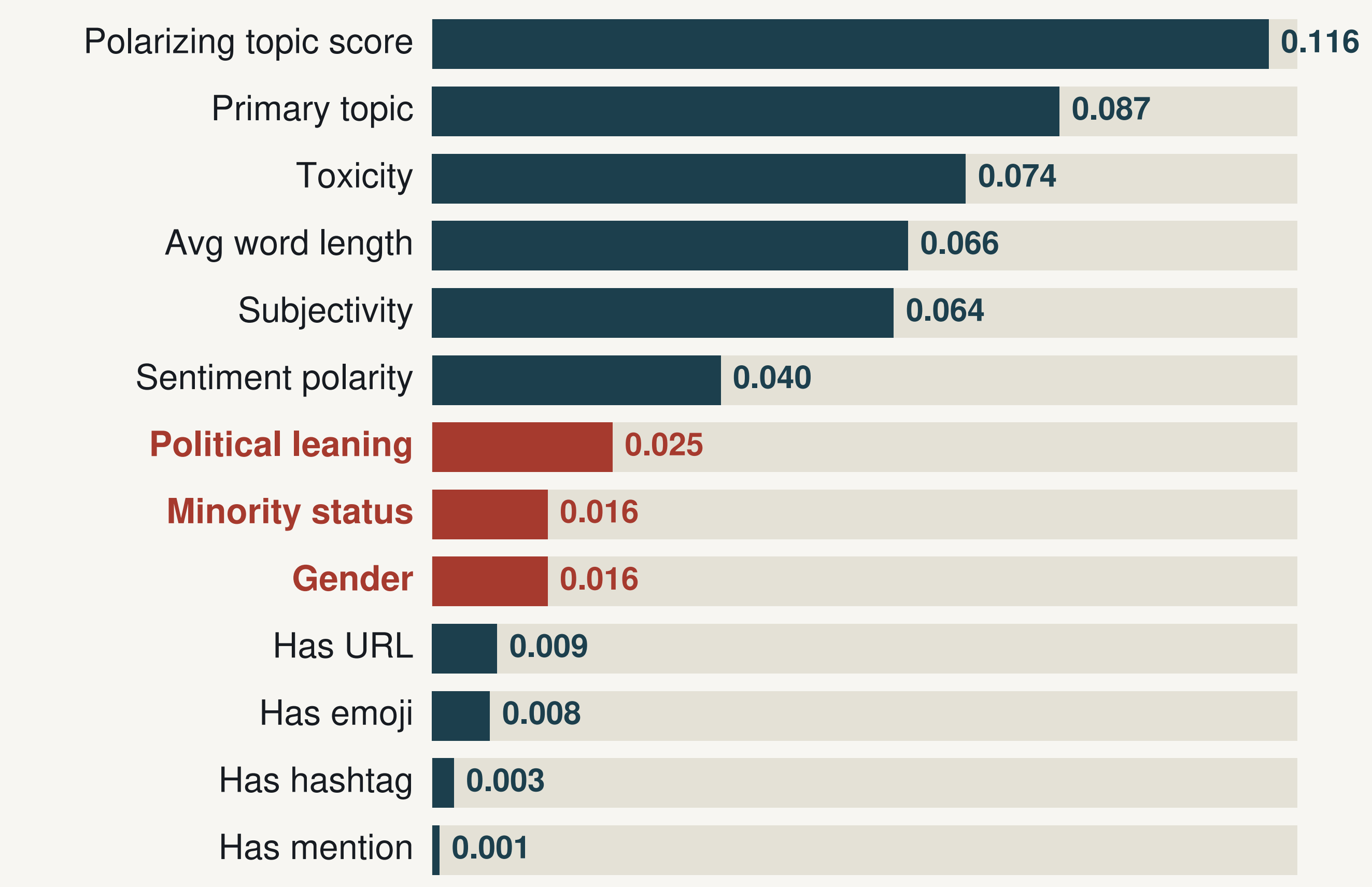
A Bias Landscape

	Controversial	Informative	Neutral	Engaging	General	Popular	Average
Polarizing topic score	.109 ***	.048 ***	.090 ***	.056 ***	.015 ***	.013 ***	.055 ***
Primary topic	.034 ***	.029 ***	.029 ***	.017 ***	.015 ***	.012 ***	.023 ***
Toxicity	.007 *	.026 ***	.013 **	.039 ***	.009 ***	.008	.017 ***
Political leaning	.025 ***	.013 ***	.018 ***	.009 ***	.006 ***	.004	.013 ***
Avg word length	.010 ***	.033 ***	.008 ***	.002 **	.005 ***	.002 *	.010 ***
Sentiment polarity	.013 ***	.001	.014	.017 ***	.003	.003 *	.009 **
Subjectivity	.003 **	.015 ***	.002 *	.003 **	.003	.003 **	.005 **
Has URL	.000	.014 **	.002	.000	.006	.000	.004
Minority status	.005 ***	.003 ***	.004 ***	.002 ***	.001	.000	.003 **
Gender	.003 ***	.004 ***	.001 **	.000	.002 **	.001 **	.002 **
Has emoji	.002 **	.002 ***	.001	.000	.000	.002 *	.001
Has hashtag	.001	.001 **	.001 *	.000	.001 **	.000	.001
Has mention	.000	.000	.000	.000	.000	.000	.000
Mean across features	.016	.015	.014	.011	.005	.004	.011

Polarizing topic score: 2× the next feature, significant in 98% of conditions.

R^2 (variance explained), averaged over providers and platforms. Red: author attributes, Twitter/X only.
** p<0.05 in >50% of conditions, ** >60%, *** >75%.*

B What Drives Selection



Mean |SHAP| of random-forest classifiers predicting selection (mean AUROC 0.746), averaged over the three providers.

C Content & Safety Directional Bias

POLARIZATION

	Claude Sonnet 4.5	GPT-4o mini	Gemini 2.0 Flash	Average
Neutral	+0.188	+0.195	+0.240	+0.208
General	+0.046	+0.086	+0.019	+0.050
Popular	-0.060	+0.066	-0.010	-0.001
Engaging	+0.199	+0.105	+0.166	+0.157
Informative	+0.139	+0.159	+0.148	+0.149
Controversial	+0.252	+0.196	+0.245	+0.231
Average	+0.127	+0.134	+0.135	+0.132

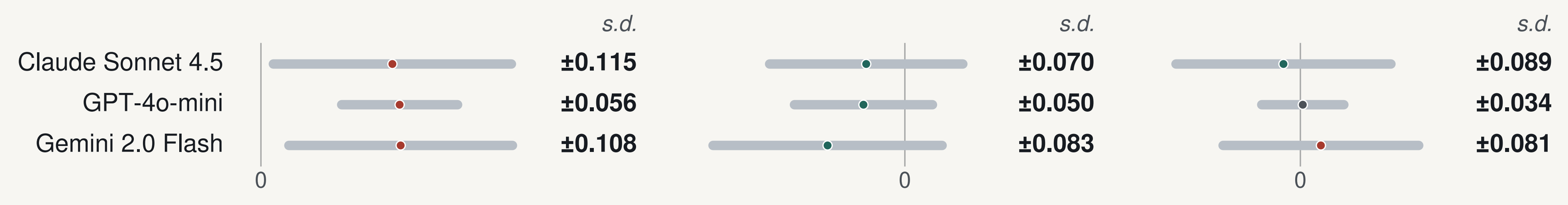
SENTIMENT

	Claude Sonnet 4.5	GPT-4o mini	Gemini 2.0 Flash	Average
Neutral	-0.007	-0.088	-0.126	-0.074
General	+0.022	+0.014	+0.029	+0.022
Popular	-0.009	+0.014	-0.013	-0.003
Engaging	-0.153	-0.027	-0.134	-0.105
Informative	+0.039	+0.003	+0.040	+0.027
Controversial	-0.059	-0.096	-0.130	-0.095
Average	-0.028	-0.030	-0.056	-0.038

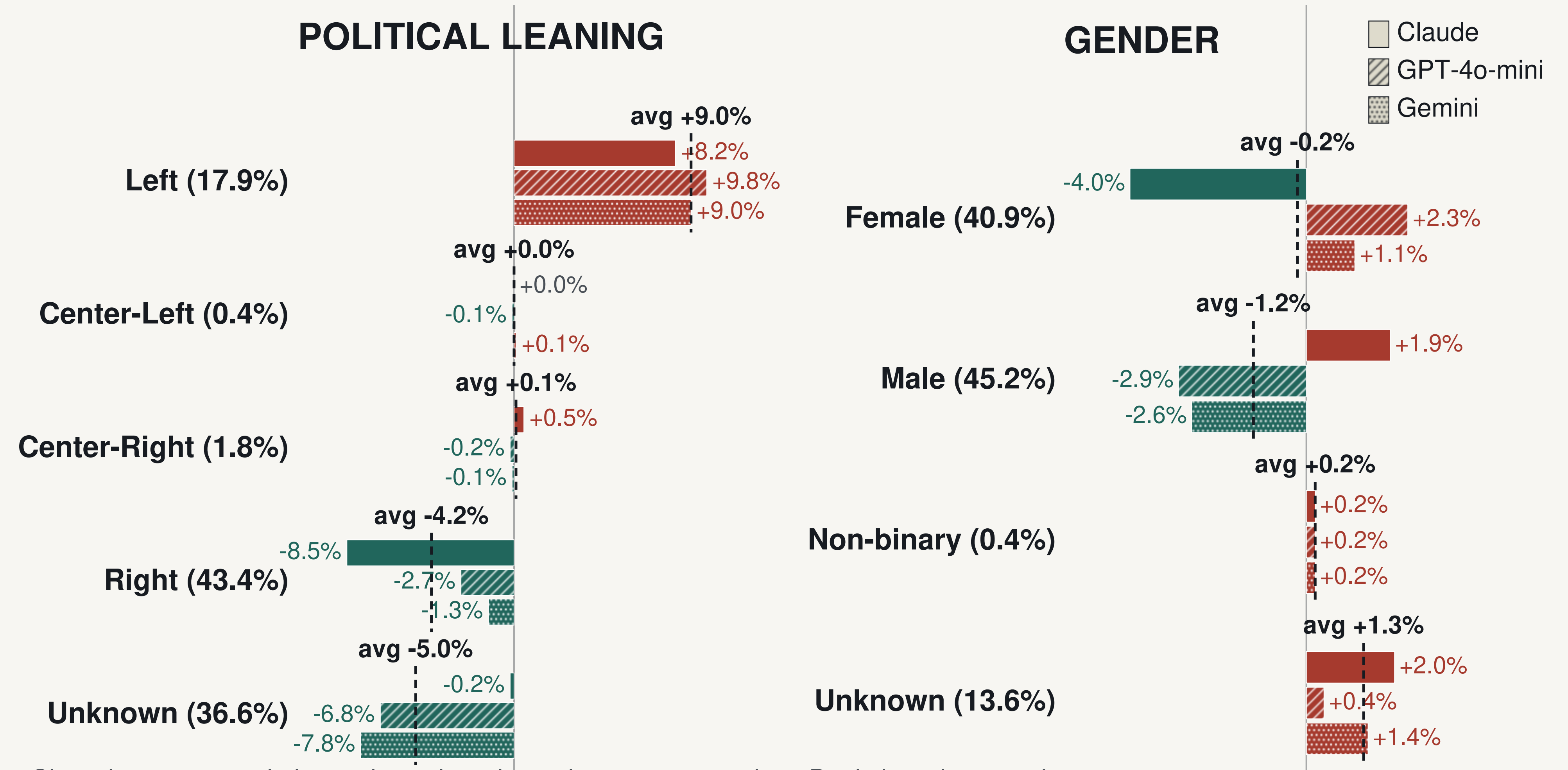
TOXICITY

	Claude Sonnet 4.5	GPT-4o mini	Gemini 2.0 Flash	Average
Neutral	-0.058	+0.034	+0.051	+0.009
General	-0.069	-0.021	-0.056	-0.048
Popular	+0.041	+0.010	+0.057	+0.036
Engaging	+0.138	+0.028	+0.120	+0.095
Informative	-0.103	-0.055	-0.101	-0.087
Controversial	-0.032	+0.017	+0.033	+0.006
Average	-0.014	+0.002	+0.017	+0.002

PROMPT SENSITIVITY *mean ± s.d. across the six prompts, per provider*

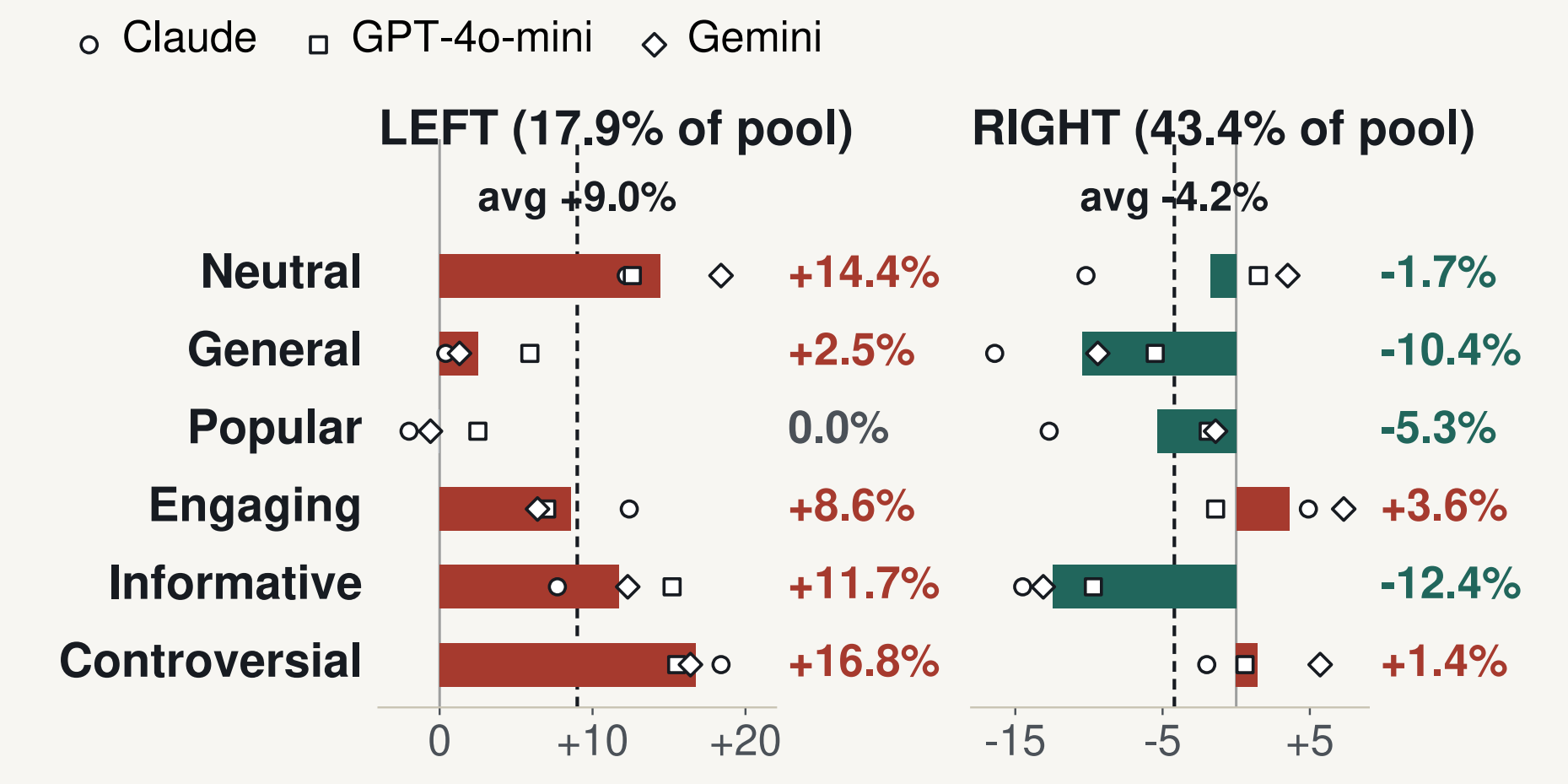


D Author Demographic Bias (Twitter/X only)



Share in recommendations minus share in pool, percentage points. Pool share in parentheses.

E Political Skew by Prompt Style



Popular prompt nearly removes the left skew.

Bar = mean across providers, markers = single providers.

TAKE-AWAY

1 **Amplification is universal.**
Recommended posts are more polarized than the pool for every provider and every platform, and under five of six prompts, including neutral.

2 **Left-leaning authors crowd the feed.**
18% of the pool, 27% of the recommendations. Right-leaning authors, the pool plurality, lose 4 points.

3 **Prompt style flips toxicity outcomes.**
Engaging prompts amplify toxicity. Informative prompts suppress it. Same models, opposite behavior.

4 **Masking demographics would not help.**
Author attributes carry little weight in the models' choices (mean SHAP 0.025 or less). The skew travels through correlated content.

Would love to stay in touch!
Nicolò Pagan · nicolo-pagan.github.io
nicolo.pagan@uzh.ch
LinkedIn: scan the code